

SATURDAY, AUGUST 15TH, 2026

OPERATIONS RESEARCH

The Transportation Problem and the Network Simplex Method

A special structure so well behaved that its linear program hands back whole numbers for free — and the reason a fifty-year-old specialised algorithm still beats general-purpose solvers on the problems that move physical goods.

2,806 words · 15-minute read

In the winter of 1941, on the frozen surface of Lake Ladoga, trucks loaded with flour and ammunition crept toward a starving Leningrad along the only supply line into a city besieged by the Wehrmacht. Drive them too close together and the ice, already groaning, would crack and swallow them. Space them too far apart and the convoy could not move enough food before the thaw or the guns found it. By one account, the man who worked out the safe spacing — as a function of ice thickness and air temperature — was a 29-year-old Leningrad mathematician named Leonid Kantorovich, then attached to the Soviet navy's engineering school [1], [2].

The story deserves a caveat: the version in which Kantorovich personally walked the route to enforce the spacing traces back to one Russian-language military history and gets embellished in retelling [1]. What is well attested is simpler, and it was not his first allocation problem. Three years earlier a Leningrad plywood trust had asked him how to split five kinds of plywood across eight lathes for maximum output [2]. That problem looks mundane next to the ice road, but in solving it Kantorovich wrote down, for the first time anywhere, the machinery now called linear programming — and specifically the version governing how to move a fixed set of goods from sources to destinations at least cost. That version has a name: the transportation problem.

Three Men, One War, One Idea

The problem was discovered independently at least three times within about three years, by people with no knowledge of one another's work, on opposite sides of a war and an ocean. Kantorovich published his production-planning paper in the USSR in July 1939 [3]. In the United States in 1941, Frank Hitchcock published "The Distribution of a Product from Several Sources to Numerous Localities," posing the same problem in almost the same language, apparently unaware of work behind the Soviet border [4]. And through the early 1940s the Dutch-American economist Tjalling Koopmans was inside Allied wartime shipping administration — bodies like the Combined Shipping Adjustment Board, deciding which ships carried which cargo out of ports on both sides of the Atlantic so as to minimize

tonnage and time lost to empty return voyages. His 1942 memorandum on optimum utilization of the transportation system, formalized in the 1951 monograph *Activity Analysis of Production and Allocation*, reached the same structure from the demands of total war [5].

Convergence from a Soviet factory floor, an American mathematics journal, and an Allied shipping desk is what happens when a civilization runs out of slack: war economies have too little of everything, so the cheapest way to move what you have to where it is needed stops being a curiosity and becomes the whole game. The machinery to answer it at scale — George Dantzig’s simplex method — arrived in 1947, and Dantzig, working at the Pentagon on logistics planning, published a version specialized to the transportation problem in 1951, in the same Koopmans-edited volume [6].

Kantorovich’s discovery, however, landed on the wrong side of an ideology. His method assigns every scarce resource an implicit price — he called these “objectively determined valuations,” today shadow prices — and those prices fall out of the mathematics whether or not anyone believes in markets. In a planned economy built on the labor theory of value, the idea that capital and land should carry a price adjusting to scarcity looked like smuggled capitalism. When he brought his methods to Gosplan in the 1940s, orthodox political economists denounced mathematical economics as an anti-Marxist import, and one reportedly invoked Vilfredo Pareto — by then associated with fascism — to discredit him by association [7], [8]. For years his published work carried prefatory apologies for its “errors,” and Kantorovich learned to write about prices without using the word [7]. In 1975 he and Koopmans shared the Nobel Memorial Prize in Economic Sciences for the theory of optimum allocation of resources [8], [9] — the only economics Nobel awarded to a Soviet citizen, and awarded, in effect, for a wartime plywood problem.

The Mathematics of Shipping Optimally

Strip away the history and the statement is simple. Suppose there are m supply points, each holding s_i units of a good, and n demand points, each requiring d_j units. Shipping one unit from source i to destination j costs c_{ij} . The decision is how much to send along each route, $x_{ij} \geq 0$:

$$\begin{aligned} &\text{minimize} && \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \\ &\text{subject to} && \sum_{j=1}^n x_{ij} = s_i \quad \forall i, \\ &&& \sum_{i=1}^m x_{ij} = d_j \quad \forall j, \quad x_{ij} \geq 0. \end{aligned}$$

This assumes total supply equals total demand, a balanced problem; any unbalanced case is repaired with a dummy source or sink at zero cost. One of the $m + n$ equality constraints is always redundant — summing the supply rows and summing the demand rows both give total throughput — so the constraint matrix has rank $m + n - 1$, which will matter twice below.

The other half of the theory is duality. Every linear program has a mirror-image dual, and here it is unusually clean. Attach a variable u_i to each source and v_j to each destination — implicit values, dollars per unit, for having goods at that source or needing them at that destination — subject to

$$u_i + v_j \leq c_{ij} \quad \text{for all } i, j.$$

Strong duality guarantees that the best such u and v exist and that the dual objective equals the primal minimum cost exactly [11]. More useful is complementary slackness: on any route carrying flow, $x_{ij} > 0$, the dual inequality holds with equality, $u_i + v_j = c_{ij}$; on an empty route it is typically strict, and the gap

$$\bar{c}_{ij} = c_{ij} - u_i - v_j \geq 0$$

is the reduced cost: the amount by which total system cost rises, per unit, if flow is forced onto a route the optimizer rejected. Those u_i and v_j are the objectively determined valuations that got Kantorovich into trouble — implicit prices emerging from the arithmetic of scarcity, with no market and no committee setting them by decree.

CHECK THIS AGAINST YOUR INTUITION

A model built to minimize exactly one number — total shipping cost — hands back a complete set of internal prices nobody asked it for. Is that a fact about linear algebra, or a fact about what "value" means under scarcity?

The two are hard to separate, which is why the result was politically explosive. Nothing in the formulation mentions prices; the dual variables are an artifact of the geometry of the feasible region. Yet they behave as prices behave, stating what an extra unit of capacity is worth at each location. A shadow price is not an ideological choice — it is what scarcity looks like in coordinates.

Why the Integers Come Free

Two structural features argue against handing this to a general-purpose solver. The first is the shape of the constraint matrix: each variable x_{ij} appears in exactly one supply row and one demand row, so every column has exactly two entries equal to 1 and the rest zero. A matrix of that form is *totally unimodular* — every square submatrix has determinant 0, +1, or −1 — and the payoff is surprising. Whenever supplies and demands are whole numbers, every optimal vertex is automatically whole-numbered too, with no integer-programming machinery required [10]. Round a generic LP's optimum to the nearest integer and the result can be far from optimal, or infeasible; the transportation problem gets integrality for free.

The second feature is what a basic feasible solution looks like. Since the matrix has rank $m + n - 1$, a basis consists of $m + n - 1$ routes, and those routes always form a spanning tree on the bipartite graph of sources and destinations. That is the engine of the transportation simplex, or stepping-stone method, that Dantzig built in 1951 [6], and of its modern generalization, the network simplex. Instead of a tableau, the algorithm maintains a tree: solve $u_i + v_j = c_{ij}$ along the tree edges to get the duals in

$O(m + n)$ arithmetic, price the non-tree routes with $\bar{c}_{ij} = c_{ij} - u_i - v_j$, and if any is negative, add that route — which creates a unique cycle — and push flow around it until one edge empties. The pivot is a graph traversal, not a matrix update. Because it moves from one integer tree solution to another it never touches a fraction, and because it factorizes nothing it avoids the numerical linear algebra that dominates general LP methods — which is why an algorithm dating to 1951 remains, on pure network problems, typically faster than the general-purpose solvers that have overtaken it everywhere else.

The same argument extends to a family of related network flow models, all of which inherit total unimodularity and therefore integrality.

Variant	Typical size	What makes it special
Transportation	m sources, n sinks	Bipartite; basis is a spanning tree of $m + n - 1$ routes
Assignment	n by n , all supplies 1	Every basic solution is a permutation matrix
Transshipment	Sources, sinks, plus relay nodes	Intermediate nodes conserve flow rather than absorb it
Min-cost flow	Any directed graph, arc capacities	Most general form; the other three are restrictions of it

Five Routes Out of Nine

Consider three crude sources — a Permian gathering system, a Bakken gathering system, and a Gulf Coast storage hub — with 40, 25, and 35 thousand barrels per day of spare deliverability, and three refineries needing 30, 45, and 25. Supply and demand balance at 100; the per-barrel costs below are illustrative, not live tariffs.

From (supply)	Refinery A	Refinery B	Refinery C
Permian (40)	4	6	8
Bakken (25)	5	4	7
Gulf Coast (35)	9	7	3
Demand	30	45	25

Nine routes are available; the optimum uses exactly five, which is $m + n - 1 = 3 + 3 - 1$, the spanning-tree count predicted above.

Route	Cost (USD/bbl)	Flow (kbbl/d)	Cost (USD k/day)
Permian to A	4	30	120
Permian to B	6	10	60
Bakken to B	4	25	100
Gulf Coast to B	7	10	70
Gulf Coast to C	3	25	75
Total	—	100	425

Total system cost is \$425,000 per day. The dual prices are $\mathbf{u} = (6, 4, 7)$ for the three sources and $\mathbf{v} = (-2, 0, -4)$ for refineries A, B, and C, normalized so that $\mathbf{v}_B = \mathbf{0}$ — because one constraint is redundant, the dual is unique only up to a constant shifted between $\mathbf{u}$ and $\mathbf{v}$. Every used route satisfies $\mathbf{u}_i + \mathbf{v}_j = \mathbf{c}_{ij}$ exactly, as duality demands: Permian to A gives $6 + (-2) = 4$, Bakken to B gives $4 + 0 = 4$, Gulf Coast to C gives $7 + (-4) = 3$.

The rejected routes are where the information is. Take Permian to Refinery C: $\mathbf{u}_1 + \mathbf{v}_3 = 6 + (-4) = 2$ against an actual cost of **8**, so $\bar{\mathbf{c}}_{13} = 6$. That gap says that diverting a barrel down that lane raises total system cost by exactly six dollars, because the diversion forces a compensating cycle of adjustments through the rest of the network — precisely the cycle the network simplex would have traversed. The other rejected lanes price out the same way: Bakken to A at $\bar{\mathbf{c}} = 3$, Bakken to C at **7**, Gulf Coast to A at **4**. All four are strictly positive, which certifies that no improving pivot exists and that this optimum is unique. Read as a schedule, the solution says which lanes run; read as a price list, it says what a marginal barrel of deliverability is worth at each source.

The Special Case That Got Its Own Algorithm

Shrink the problem so that every source supplies exactly one unit and every destination demands one — one worker to one job, one aircraft to one gate, one donor organ to one recipient — and the result is the assignment problem: the one-to-one pairing of minimum total cost. It is mathematically a transportation problem, but its bipartite matching structure earned a dedicated method: the Hungarian algorithm, published by Harold Kuhn in 1955 and built on 1930s results of Dénes Kőnig and Jenő Egerváry, hence the name [12]. In 2006 researchers found that Carl Gustav Jacobi had in effect already solved it, in a manuscript published posthumously in Latin in 1890 — sixty years in an archive before Kuhn rediscovered its logic [13]. James Munkres proved in 1957 that the method is strongly polynomial, and refinements by Jack Edmonds and Richard Karp brought it to the now-standard order $\mathbf{n}^3$ [13].

That efficiency is the point, because the naive alternative — enumerating every pairing — grows as $\mathbf{n}!$. For twenty workers and twenty jobs that is more than 2×10^{18} permutations, beyond what any desktop could enumerate in a lifetime, against a Hungarian solve that finishes in microseconds. The gap between $\mathbf{n}!$ and $\mathbf{n}^3$ comes entirely from exploiting structure a general-purpose method ignores.

From Dirt Piles to Neural Networks

The problem's ancestor is older than Kantorovich or Hitchcock. In 1781 Gaspard Monge posed what he called *déblais et remblais*, excavation and fill: given a pile of earth in one place and a hole to fill in another, what pattern of moving each shovelful minimizes total work, measured as mass moved times distance moved [14]? Monge required each infinitesimal bit of earth to travel whole to a single destination — a rigid assignment, not a blend — which makes the problem nonconvex and, for two centuries, intractable. Kantorovich's second insight, published in 1942 as "On the Translocation of Masses," was to relax that rule and let mass split across destinations, turning a rigid assignment into a probability distribution over routes [15] — the same move as in the crude example, generalized from discrete barrels to continuous distributions. The resulting minimum cost between two distributions is now called the Wasserstein distance.

TRY THIS BEFORE READING ON

The discrete shipping problem, Monge's dirt piles, and a distance between probability distributions look like three unrelated subjects. What is the single object that all three are instances of, and what changes between them?

What changes is only the measure being moved: barrels, mass, probability. The object is the same — a minimum-cost coupling between two distributions over a space with a cost function — and each level inherits the duality of the one below. Where duality gave 1940s economists implicit prices, it gives machine learning a distance between distributions that behaves sensibly even when they do not overlap, a property more familiar measures like KL-divergence lack. The 2017 Wasserstein GAN paper by Martin Arjovsky, Soumith Chintala, and Léon Bottou replaced a GAN's notoriously unstable objective with an estimate of this optimal-transport distance and obtained dramatically more reliable training [16]. Optimal transport now turns up in comparisons of single-cell genomic profiles, in matching color palettes between images, and in estimating how far a climate model's rainfall distribution sits from the observed one [17] — each still asking Hitchcock's 1941 question: what is the cheapest way to move this distribution of stuff onto that one?

The Unfinished Argument

A quiet irony sits at the center of the story. The mathematics does not care whether the prices it generates come from a Politburo's five-year plan or a trading floor — they are the same object either way, an artifact of scarcity and arithmetic rather than of any economic system. Kantorovich spent three decades arguing that this made his mathematics safely socialist. Free-market economists in the West cited the same duality theorem as evidence that markets were mathematically inevitable, since decentralized price-taking reproduces the optimum a planner with perfect information would compute

by solving the linear program directly [8]. Both readings cannot be the whole truth, and the disagreement was overtaken rather than resolved: the state that might have used his methods at full scale dissolved a decade and a half after his Nobel Prize.

What survives is the structure, indifferent to the argument it was drafted into — a plywood-scheduling trick from 1939 that describes with equal fidelity how to ration food across a besieged city, how to price a barrel of crude, and how to train a network to generate images. It also leaves a standing obligation. The word “optimal” in a solver’s output is a claim about one objective function and one set of constraints, and the reduced costs are the record of everything that claim ruled out. When software reports the optimal allocation of something scarce, the useful question is not whether the number is right but what price it just assigned to everything it declined to do, and whether anyone has read that list.

REFERENCES

- [1] V. D. Dotsenko, *Флот. Война. Победа. 1941–1945*; excerpt and sourcing discussion in “Can somebody help me track down the source claiming Leonid Kantorovich was responsible for safety on the Road of Life during the Siege of Leningrad?,” AskMetaFilter, 2013. Secondary discussion of a Russian-language primary account; popularized retellings, e.g. Kantorovich personally walking the convoy route, are not corroborated in the underlying source.
- [2] J. J. O’Connor and E. F. Robertson, “Leonid Vitalievich Kantorovich,” MacTutor History of Mathematics Archive, University of St Andrews.
- [3] L. V. Kantorovich, “Mathematical Methods of Organizing and Planning Production,” Leningrad State University, 1939; trans. *Management Science*, vol. 6, no. 4, pp. 366–422, 1960.
- [4] F. L. Hitchcock, “The Distribution of a Product from Several Sources to Numerous Localities,” *Journal of Mathematics and Physics*, vol. 20, pp. 224–230, 1941, doi:10.1002/sapm1941201224.
- [5] T. C. Koopmans, “Optimum Utilization of the Transportation System,” in *Activity Analysis of Production and Allocation*, Cowles Commission Monograph 13, T. C. Koopmans, Ed. New York: Wiley, 1951. See also “Combined Shipping Adjustment Board,” for institutional context on Allied wartime shipping coordination.
- [6] G. B. Dantzig, “Application of the Simplex Method to a Transportation Problem,” in *Activity Analysis of Production and Allocation*, T. C. Koopmans, Ed. New York: Wiley, 1951, pp. 359–373.
- [7] “Programming the USSR: Leonid V. Kantorovich in Context,” *British Journal for the History of Science*, Cambridge University Press.
- [8] INFORMS, “Leonid V. Kantorovich,” History of O.R. Excellence, Biographical Profiles.
- [9] “The Sveriges Riksbank Prize in Economic Sciences in Memory of Alfred Nobel 1975 — Press Release,” Nobel Prize Foundation.
- [10] Lecture notes on totally unimodular matrices and network flow polytopes, standard combinatorial optimization course material (e.g. the Stanford CS/MS&E theory sequence); see also transportation-simplex treatments citing Dantzig [6].
- [11] “Duality in Linear Programming,” MIT 15.053 course notes, Massachusetts Institute of Technology.
- [12] H. W. Kuhn, “The Hungarian Method for the Assignment Problem,” *Naval Research Logistics Quarterly*, vol. 2, pp. 83–97, 1955.
- [13] “Hungarian algorithm,” Wikipedia, for the 2006 discovery of Jacobi’s 1890 posthumous solution and the Munkres 1957 and Edmonds–Karp complexity results.
- [14] G. Monge, “Mémoire sur la théorie des déblais et des remblais,” *Histoire de l’Académie Royale des Sciences*, 1781; discussed in C. Villani, *Optimal Transport: Old and New*. Berlin: Springer, 2009.
- [15] L. V. Kantorovich, “On the Translocation of Masses,” 1942; reprinted in *Management Science*, vol. 5, no. 1, pp. 1–4, 1958, doi:10.1287/mnsc.5.1.1.
- [16] M. Arjovsky, S. Chintala, and L. Bottou, “Wasserstein GAN,” arXiv:1701.07875, 2017.

[17] “A Survey on Algorithmic Developments in Optimal Transport Problem with Applications,” arXiv:2501.06247, 2025.

Daily Learning No. 003 · Operations Research · Saturday, August 15th, 2026 · Sources are listed above; every load-bearing figure traces to a numbered reference.